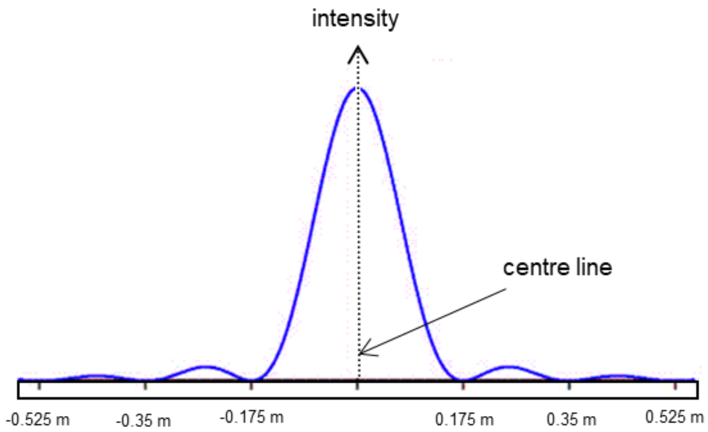
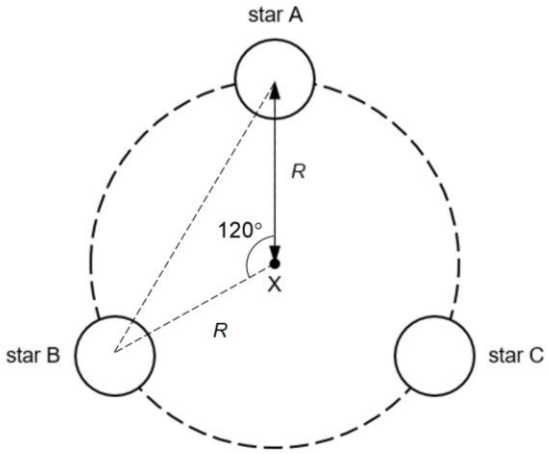
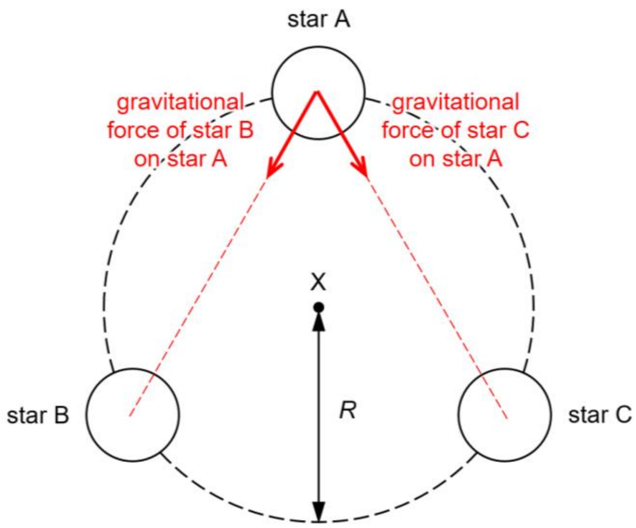
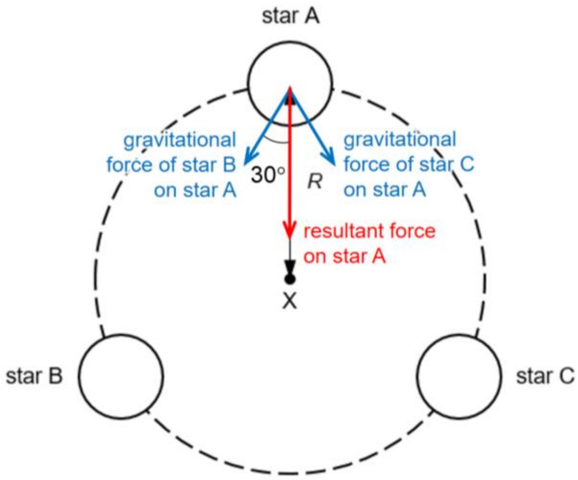
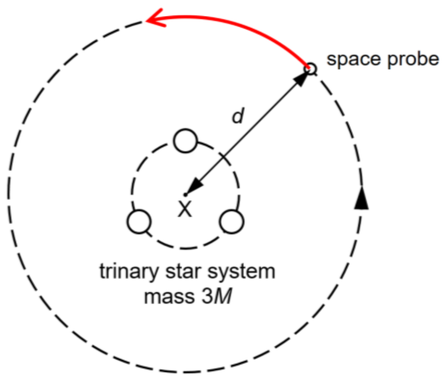
5(a)	The principle of superposition states that the net displacement at a given place and time caused by two or more waves which transverse the same space and meet is.
	the vector sum of the displacement which would have been produced by the individual waves separately at that position and instant of time
(b)(i)	$x = \frac{\lambda D}{a}$ $= \frac{(650 \times 10^{-9})(6.5)}{(1.65 \times 10^{-3})}$
	$x = 2.56 \times 10^{-3} m$
(ii)	Bright fringe becomes less bright as the amplitude of the light from the darken slit is lower
	Dark fringe will increase in brightness as the two wave cannot fully cancel each other
(iii)	$\theta_R = \frac{\lambda}{b}$ $= \frac{650 \times 10^{-9}}{3.5 \times 10^{-3}}$ $= 1.86 \times 10^{-4} rad$
	$d \approx r\theta$ $1.65 \times 10^{-3} = r(1.86 \times 10^{-4})$ $r = 8.88m$
(iv)	When pupil becomes larger, θ_R becomes smaller
	Therefore he will be able to resolve the 2 sources from an even further distance. He will still be able to resolve the 2 light sources

(c)(i)	$\sin\theta = \frac{\lambda}{b}$ $= \frac{550\times10^{-9}}{2.20\times10^{-6}} = 0.25$
	$\sin\theta = \frac{dist}{0.7} = 0.25$ $dist = 0.175$ $width = 2(0.175) = 0.35m$
ii)	
	Symmetrical, intensity of centre bright fringe significantly higher (at least 1:5) than the adjacent fringe
	Width of centre bright fringe twice of other fringes, width properly labelled
iii)	$I = \frac{P}{S} = \frac{P}{4\pi r^2}$ $I = kA^2$ $A \alpha \frac{1}{r}$
	$\frac{A_{new}}{A_0} = \frac{r_0}{r_{new}}$ $A_{new} = \frac{0.8}{1.1}A_0 = 0.727A_0$

7	
(a)	The direction of the object's motion changes. Therefore, its velocity changes and it accelerates.
	Since it travels at a constant speed, the direction of the acceleration, and hence resultant force is perpendicular to the direction of motion.
(b)(i)	 distance between stars $= \sqrt{R^2 + R^2 - 2(R)(R)\cos(120°)}$ $= R\sqrt{2 - 2\cos(120°)}$ $= 1.732R$
	$= 1.73R$
(b)(ii)	
	The two forces are drawn with equal magnitudes and in the correct directions.
	The two forces are labelled correctly.

(b)(iii)1.	gravitational potential energy $= U_{AB} + U_{BC} + U_{AC}$
	$= -\dfrac{GM^2}{1.73R} + \left(-\dfrac{GM^2}{1.73R}\right) + \left(-\dfrac{GM^2}{1.73R}\right)$ $= -1.73\dfrac{GM^2}{R}$
(b)(iii)2.	Consider star A 
	gravitational force of one star on another star $= \dfrac{GM^2}{(1.73^2)R^2}$ $= 0.333\dfrac{GM^2}{R^2}$
	resultant force on each star $= \dfrac{GM^2}{3R^2}\cos(30°) + \dfrac{GM^2}{3R^2}\cos(30°)$ $= 0.577\dfrac{GM^2}{R^2}$
(b)(iii)3.	The resultant gravitational force on each star provides for the centripetal force
	$0.577\dfrac{GM^2}{R^2} = \dfrac{Mv^2}{R}$ $\dfrac{0.577}{2}\dfrac{GM^2}{R} = \dfrac{1}{2}Mv^2$ kinetic energy of one star $= 0.289\dfrac{GM^2}{R}$ kinetic energy of trinary star system $= 3 \times 0.289\dfrac{GM^2}{R}$ $= 0.866\dfrac{GM^2}{R}$

(b)(iv)	The resultant force acting on each star is not sufficient to provide for the centripetal force to keep the stars in orbit.
	The stars will spiral outwards into an orbit with a larger radius.
(c)(i)1.	The space probe can observe the same part of the trinary star system.
(c)(i)2.	The space probe can observe different parts of the trinary star system.
(c)(i)3.	The same side of the space probe always faces the trinary star system.
(c)(ii)	
(c)(iii)	To escape the gravitational field of the trinary star system, the total energy of the space probe of mass *m* must be minimally equal to zero
	$\frac{1}{2}mv^2+\left[-\frac{G(3M)m}{d}\right]\geq 0$ $v^2\geq\frac{2G(3M)}{d}$ $v\geq\sqrt{\frac{2G(3M)}{d}}$ $v\geq\sqrt{\frac{2(6.67\times10^{-11})(3\times1.39\times10^{30})}{1.05\times10^{11}}}$ $v\geq 7.278\times10^{4}$ m s^{-1}
	$v=7.28\times10^{4}$ m s^{-1}